

(For the Group Submission, follow [this template](#) instead)

Design Contributions Summary

Individual Contributor's Name: *Justin Kirkland*

Contributions to *Room 1: MP1b*

Team Number: 7

Team Members: *Kyle Li, Anshi Mathur*

Github Repository Link: *github.com/kyle-li5/cs417-mp1/tree/justin-mp1b*

Individual itch.io Build Link: *jk4212.itch.io/cs-417-mp1b*

Feature Stories Contributed

List all the feature stories you believe you have completed from the open-ended game design rubric. Explicitly state the name of that feature and then describe what elements visible in the submitted video correspond to that particular feature. At the end of the list item, include a timestamp for a time when it can be seen. For feature stories that allow multiple instances (e.g. juicy feedback), include a bulleted sublist describing each instance and how it can be seen in the video. For example:

Simulation World (W) Stories

1. Key Props with Grab Affordances (1 points):

There are at least 3 objects with Grab Interactables:

- a. Battery (XRGrabInteractable)
- b. Fuse (XRGrabInteractable)
- c. Breakers (XRGrabInteractable)

2. Solid Objects (1 points):

- a. All keys collide, can fall, can be thrown, floor has Collider

3. Locks with Accepting Affordances (3 points):

At least 3 unlocking events defined by a Key Prop entering a Collider

- a. Battery enters battery terminal (XRSocketInteractor)
- b. Fuse enters fuse slot on breaker box (XRSocketInteractor)
- c. Breakers enter their respective slots in breaker box (XRSocketInteractor)

Execution (X) Stories

1. Escaping the Room (3 points):

The player must trigger three Lock scripts to progress through the game towards escaping and win.

- a. Stage 0: Circuit puzzle and breaker box's functionality are disabled entirely until the battery is inserted into the terminal
- b. Stage 1: Battery is inserted, circuit puzzle activates but breaker box remains functionless
- c. Stage 2: Battery is inserted, circuit puzzle is solved, => breaker box's functionality is enabled

All three puzzles must be sequentially solved before the exit door slides open

2. Signifiers

- a. Grab Signifiers (2 points): Interactable objects are textured with a "glowing" yellow material
- b. Escape Signifiers (2 points): Pipes are used to direct the player's attention to/between the various stages of the room. The pipes fade from gray to green to signify the flow of electricity from previous stages on activation of a key/lock combo.
- c. Lock Signifiers (4 points):
 - i. The battery (key) / battery terminal (lock) pair are made apparent to the player via the familiar positive/negative contact symbols on the battery terminal colored red/green respectively.
 - ii. The fuse (key) / fuse slot (lock) combo is made apparent to the player by an equally sized slot on the side of the breaker box and a hint revealed when the breaker box's door is opened. In addition, a main indicator turns from red to green only when all breakers and fuse are installed.
 - iii. The breakers (key) / breaker box (lock) is made apparent by a corresponding indicator light and label of the respective slot. The indicator light for a particular slot is enabled only when a breaker is installed. These indicators indicate the breaker's state (on/off) and combined with the labels guide the user to the correct on/off combination that solves the puzzle.
- d. Repetition & Variety in Lock Mechanics (extends Lock Signifiers) (2 points): Each key/lock combo is unique but unified in theme. The overall theme of all three combos is electricity flow: first electricity flows from the battery/battery terminal to the circuit puzzle. Then the tiles of the puzzle

must be shifted to route the electricity input to the output on the right-hand side. Finally, the breaker box must be configured to route the electricity passed by the circuit puzzle to the exit door.

3. User Feedback (U) Stories:

- a. Eased State Changes (2 points): The pipes to/from each component ease from gray to green to indicate the flow of electricity is established. This directs the attention of the player to the next puzzle.

Sidequests

Unlocks:

- A. Reveals (2 points): The fuse falls out of its container when the circuit puzzle is solved, revealing it to the player by removing a bottom panel.

Minigames

- A. Puzzle System (2 points): The circuit puzzle is a separate stand-alone puzzle where the player has to physically rearrange the tiles by interacting with them. The state is eased in/out with animations.
- B. Puzzle Content (extends Puzzle System): Overall, I have two puzzle minigames for +1 point: the circuit puzzle and the breaker boolean combination puzzle.
- C. Puzzle Discoverability (extends Puzzle System) (3 points): The slot labels combined with the hint on the backside of the door panel guide the player to the correct configuration

Claimed points: 28/33 points

- A. Critical Path: 20/20 points
- B. Sidequests: 8/10 points

Note to grader:

After many hours in the 222 Siebel lab, I experienced a partial corruption of my room early Thursday, the day of the original individual+team deadline. Best as I was able to trace, this resulted from a circular dependency inadvertently introduced during development leading to an infinite compilation loop in Unity. Any attempt in reloading the scene resulted in a similar infinite loading loop.

While I was able to retain my pre-existing code, I lost all of my scene progress and all of my prefabs. Thus, this greatly hampered my ability to meet the 1-day individual contribution extension of Friday as I both had to fully remake lost assets and continue development of planned features. If permissible, any possible leniency regarding the late-grade policy of 50% credit would be greatly appreciated.

Thank you so much for your time and consideration!